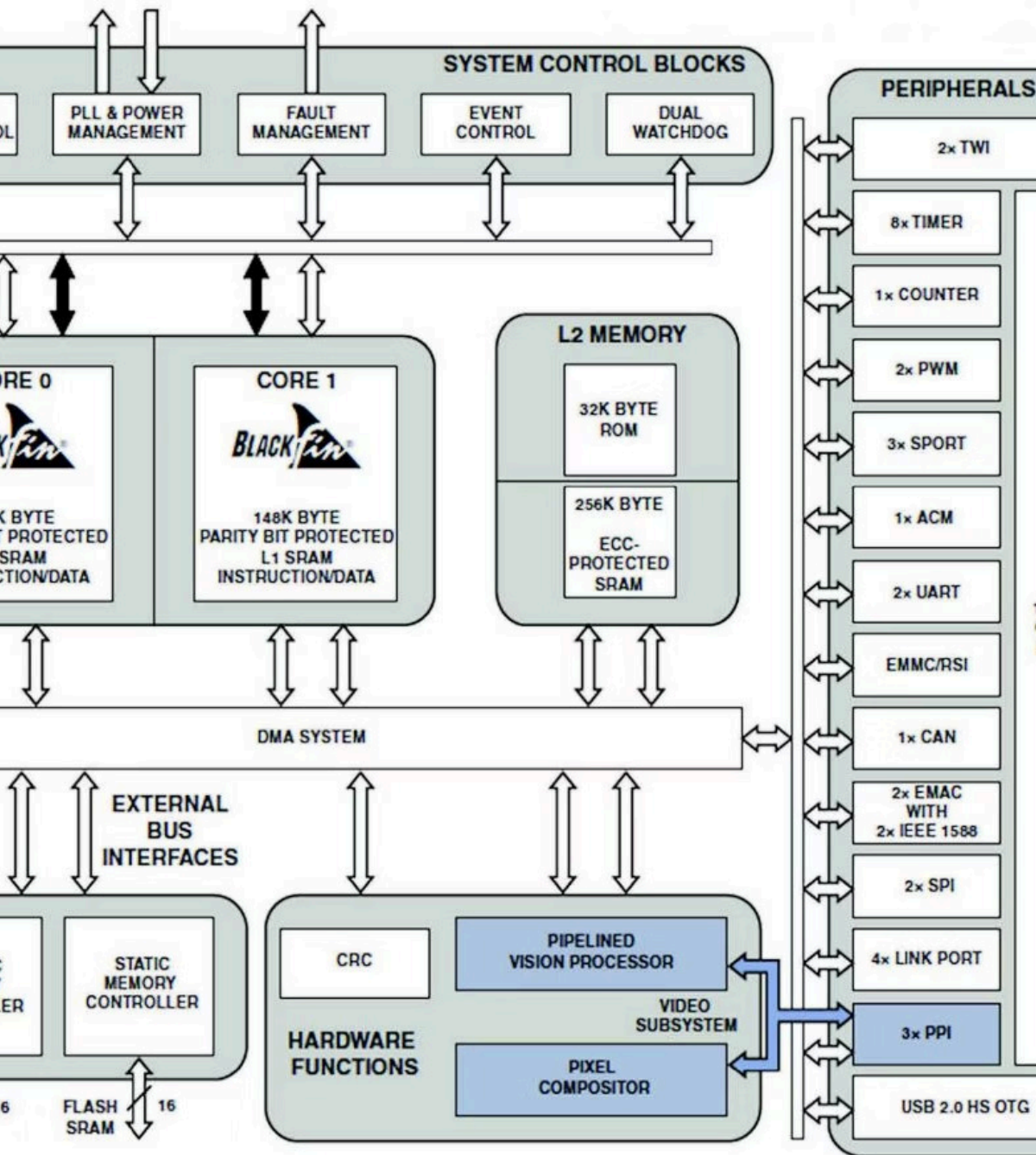
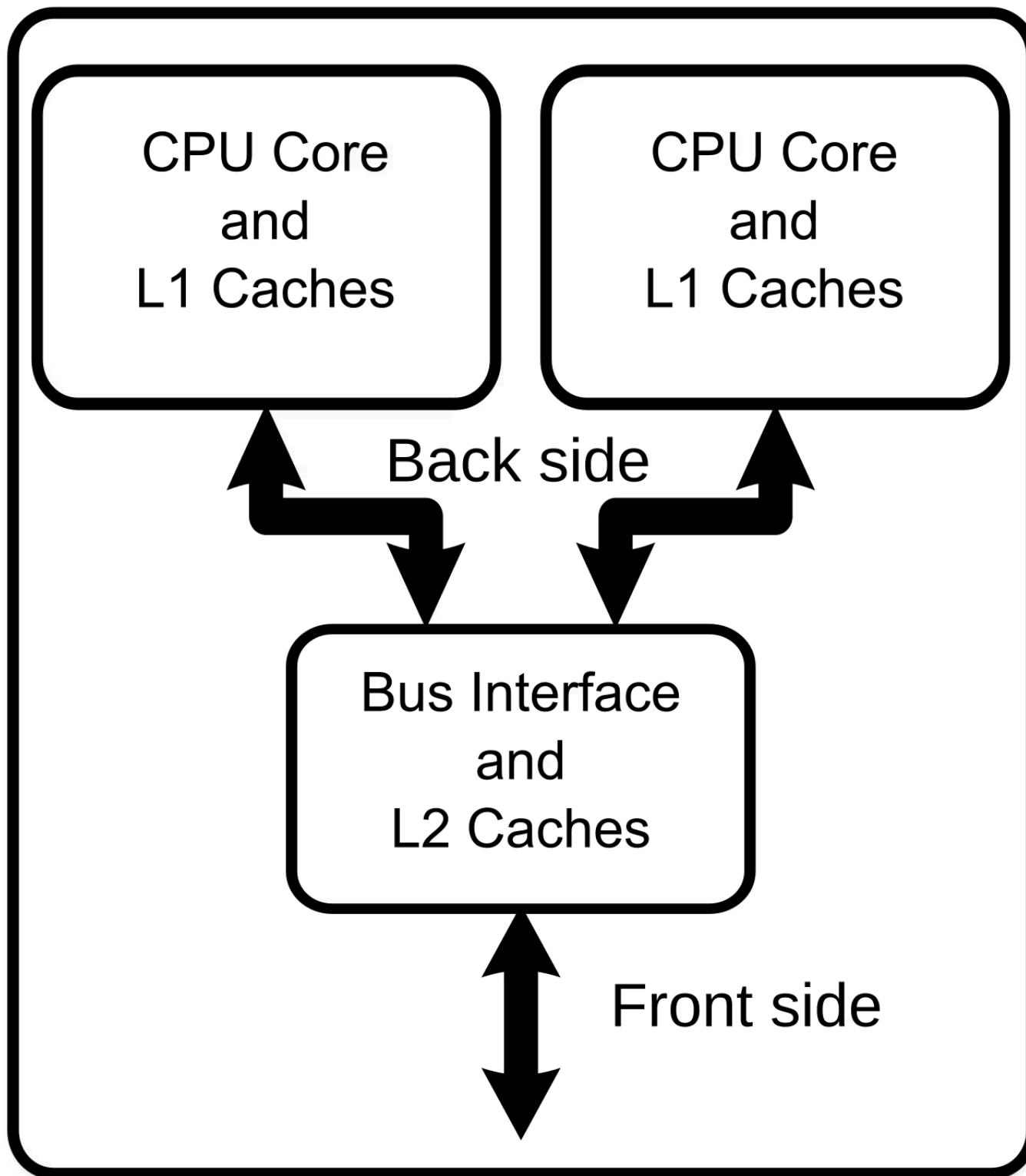




The Blackfin BF609 Powering Embedded Vision

Analog Devices' high-performance dual-core processor built for the demands of intelligent, low-power embedded systems.

Introduction: A Dual-Core Powerhouse



The **Blackfin BF609** from Analog Devices is a high-performance dual-core processor purpose-built for embedded vision applications. It uniquely combines **DSP signal processing** and **microcontroller control capabilities** on a single efficient architecture.

Dual-Core

Two independent cores for parallel processing

Low Power

Optimized for thermally constrained designs

Vision-Ready

Hardware-accelerated image processing built-in

Architecture: The Heart of Blackfin



Dual-Core Symmetric Design

Two high-performance cores running up to **500 MHz** each, enabling true parallel processing for compute-intensive vision tasks.



RISC-Like Instruction Model

Simplified instruction set improves compiler efficiency, reduces development time, and accelerates time-to-market.



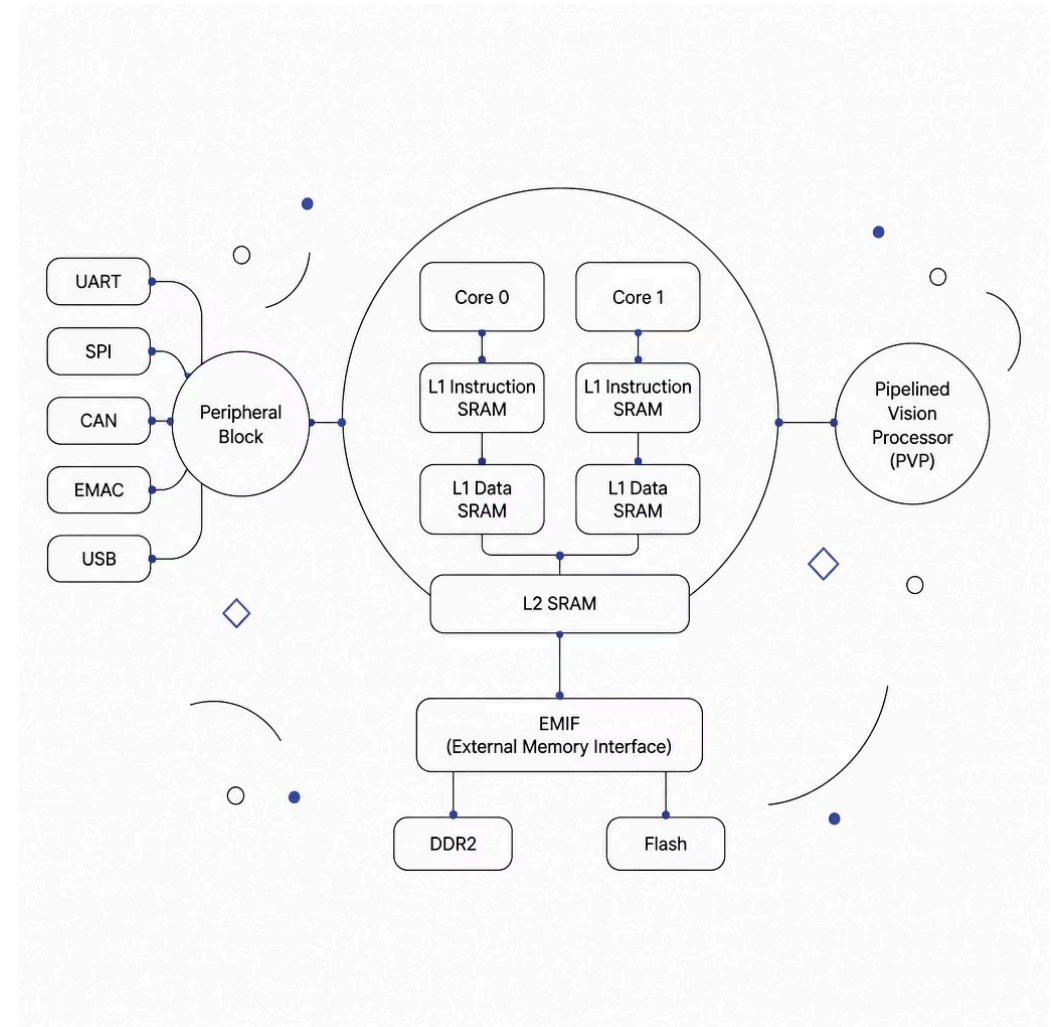
Advanced Debug & Monitoring

Integrated trace, performance counters, and real-time debugging tools streamline development and optimization.

Block Diagram: A Visual Overview

The BF609 integrates all major system components on-chip, minimizing external dependencies. Key blocks include:

- **Dual Blackfin Cores** — up to 500 MHz each
- **L1/L2 SRAM** — low-latency on-chip memory
- **PVP** — Pipelined Vision Processor for image acceleration
- **EMIF** — External Memory Interface for DDR2/Flash
- **Peripherals** — UART, SPI, CAN, EMAC, USB 2.0



Working Principle: DSP Meets Microcontroller

Each Blackfin core is a unified engine combining signal processing and control in a single architecture.



Dual 16-bit MACs

Two Multiply-Accumulate units per core deliver high-throughput DSP operations for filtering, transforms, and convolution.



Dual 40-bit ALUs

Two 40-bit Arithmetic Logic Units handle complex integer and fractional calculations with single-cycle execution.



40-bit Barrel Shifter

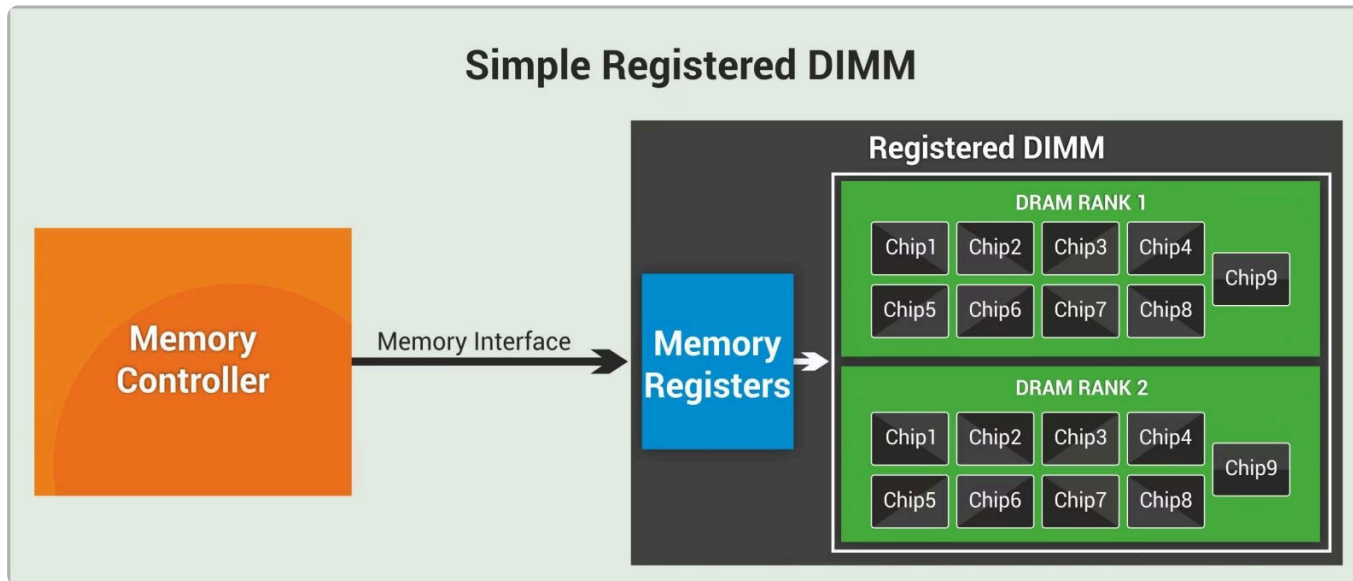
Enables efficient bit-level data manipulation, scaling, and normalization in a single clock cycle.



Pipelined Vision Processor

Dedicated hardware accelerator offloads image pre-processing and co-processing tasks from the main cores.

EMIF Interface Architecture



The **External Memory Interface (EMIF)** provides seamless connectivity to a wide range of external memory devices, enabling flexible and scalable system designs.

Dynamic Memory Controller

16-bit interface supporting **DDR2** and **LPDDR** DRAM — essential for high-bandwidth image buffers.

Static Memory Controller

8-bit and 16-bit asynchronous interface for **NOR Flash**, **NAND Flash**, and **SRAM** devices.

EMIF: Speed, Flexibility & Vision Optimization



High Bandwidth

Supports rapid data transfer rates critical for real-time video frame buffering and image processing pipelines.



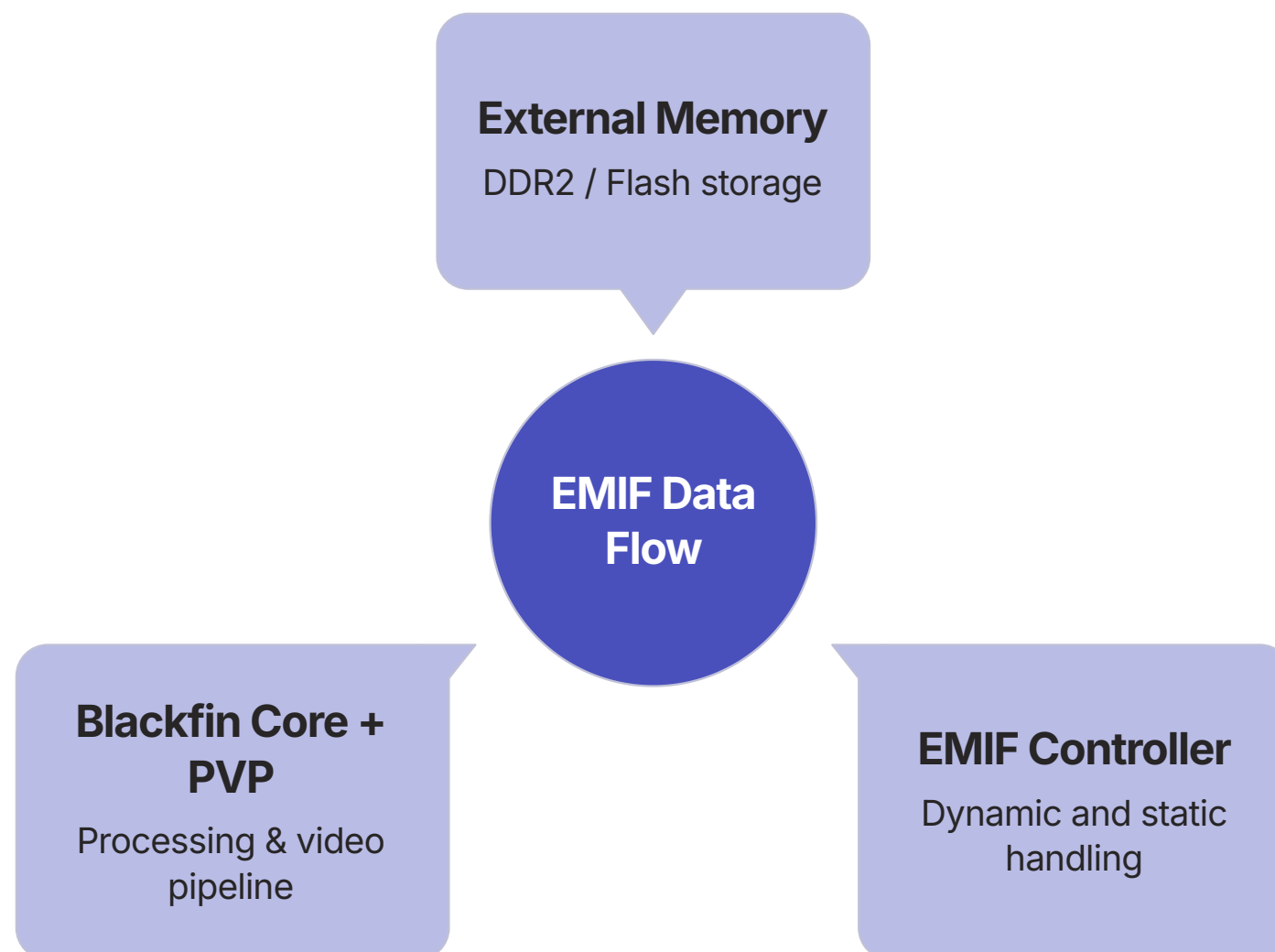
Versatile Memory Support

Accommodates DDR2, LPDDR, Flash, and SRAM in multiple configurations — reducing BOM complexity.



Vision-Optimized

Designed to handle large, continuous image and video data streams without bottlenecks.



Data flows seamlessly from external memory through the EMIF controller into the processing cores.

Beyond the Core: Standout Features

1 Pipelined Vision Processor (PVP)

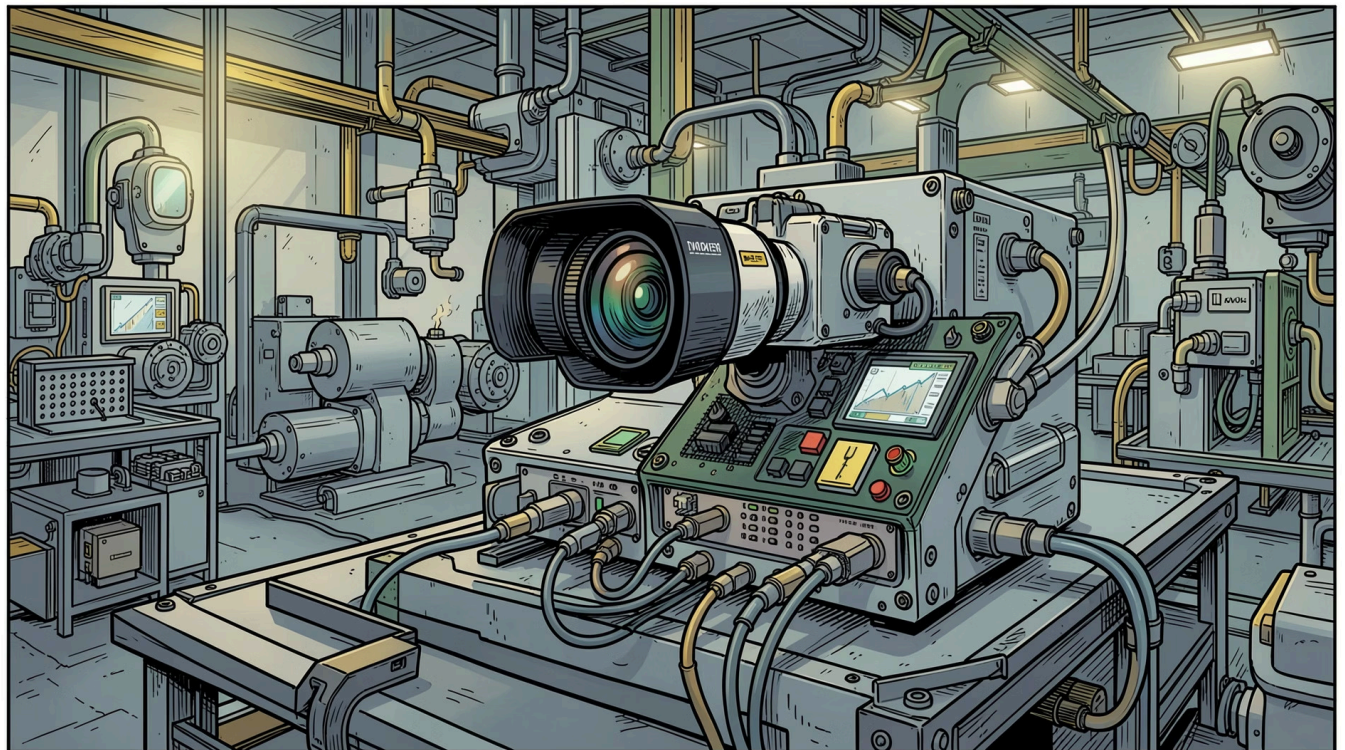
Hardware acceleration for video pre-processing — color space conversion, scaling, and filtering — offloading the CPU cores.

2 Flexible Boot Options

Boots from SPI Flash, SD/eMMC, parallel Flash, or host interfaces — enabling diverse deployment scenarios.

3 Rich Peripheral Set

UART, SPI, I²C, CAN, EMAC (Ethernet), SPORT, and USB 2.0 HS OTG for comprehensive system connectivity.



Why BF609? The Competitive Advantage

500MHz

Core Speed

Per core for real-time processing

2

Independent Cores

Symmetric dual-core design

16

Bit MAC Units

Dual MACs per core

Low Power Consumption

Ideal for battery-powered and thermally constrained embedded devices without sacrificing performance.

Integrated Vision Processing

On-chip PVP reduces external component count, lowering BOM cost and system complexity.

Robust Ecosystem

Comprehensive toolchain, SDK, and support from Analog Devices accelerates development cycles.

The Future is Vision: BF609 in Action

The Blackfin BF609 is empowering the next generation of intelligent systems across industries.

ADAS

Advanced driver assistance with real-time lane detection and object recognition

Industrial Automation

Machine vision for quality inspection, robotics guidance, and process control

Intelligent Surveillance

Edge-based video analytics for security and smart city applications

- ❏ The BF609 represents Analog Devices' commitment to high-performance, low-power embedded processing — bridging the gap between DSP performance and microcontroller simplicity.